Sensors and Actuators A 265 (2017) 261–274

Contents lists available at ScienceDirect

Sensors and Actuators A: Physical

journal homepage: www.elsevier.com/locate/sna

Vaporizing Liquid Microthrusters with integrated heaters and temperature measurement

Marsil A.C. Silva [a,*], Daduí C. Guerrieri [a], Henk van Zeijl [b], Angelo Cervone [a], Eberhard Gill [a]

[a] Faculty of Aerospace Engineering, TU Delft, The Netherlands
[b] Else Kooi Laboratory, TU Delft, The Netherlands

ARTICLE INFO

Article history:
Received 14 April 2017
Received in revised form 14 July 2017
Accepted 17 July 2017
Available online 5 September 2017

Keywords:
Microresistojet
Microthruster
MEMS

ABSTRACT

This paper presents the results of design, manufacturing and characterization of Vaporizing Liquid Microthrusters (VLM) with integrated molybdenum heaters and temperature sensing. The thrusters use water as the propellant and are designed for use in CubeSats and PocketQubes. The devices are manufactured using silicon based MEMS (Micro Electro Mechanical Systems) technology and include resistive heaters to vaporize the propellant. The measurements of the heaters' resistances are used to estimate the temperature in the vaporizing chamber. The manufacturing process is described as well as the characterization of the thrusters' structural and electrical elements. In total 12 devices with different combinations of heaters and nozzles have been assessed and four of them have been used to demonstrate the successful operation of the thrusters. Results are used to validate the thrusters and show a performance close to the design parameters and comparable to other devices found in the literature.

1. Introduction

Nano- and pico-satellites have grown in popularity in the last decades as these spacecraft have been applied in a wide range from industrial to educational applications. They have also been used to demonstrate novel technologies and as tools especially for Earth observation [1–3]. The maturity of this class of satellites is high but the research on micropropulsion for these spacecraft is still at an early stage and the systems are not yet at the required maturity.

Micropropulsion systems can be applied for a variety of functions, such as the execution of precise orbital and attitude maneuvers which are important for the execution of missions such as space debris removal, orbit transfer, and formation flying. Recently, many different micropropulsion systems have been developed particularly focused on the implementation on CubeSats [4].

Within micropropulsion systems, microresistojets are a very interesting choice, specifically for CubeSats, since it is one of the few types of such systems currently able to achieve a thrust level in the range 0.1–10 mN while still meeting all the constraints posed by extremely miniaturized spacecraft. The principle of microresistojets is based on heating a gaseous propellant with a resistance and then accelerating and expelling it to space. Some devices use propellants stored in liquid or solid phase, hence a phase-change process is required prior to the heating of the gas. The phase-change is done by heating a resistance in contact with the propellant that is kept in certain conditions of pressure and temperature to allow the phase change (sublimation or vaporization) to occur. Devices that use liquid propellants are called Vaporizing Liquid Microthrusters (VLM) and has been investigated by different research groups as will be discussed in the sequel.

Different designs can be found in the literature reporting the development of microresistojets. In [5] a micro-thruster is presented and tested; it consists of layers of ceramic. The micro-thruster is built by combining three layers of ceramic material: the combustion chamber, the inlet channel, and the nozzle are cut in the inner layer and a micro-heater is attached to the third layer. Tests were performed using water as propellant and it was found that the ceramic thruster is slightly more efficient than some silicon thrusters with respect to power consumption and delivered thrust. Authors in [6] present the details of fabrication and test of a low temperature co-fired ceramic (LTCC) micro-thruster. They analyze the results for pressure, temperature, power and thrust and also present some comments about the relation between the temperature of the chamber and the vaporization of the propellant.

In [7] the design, simulation, fabrication and test of a VLM are presented. The design of the chamber is based on basic calculations with temperature and residence time. The inlet channel is designed

* Corresponding author.
E-mail address: m.deathaydecostaesilva@tudelft.nl (M.A.C. Silva).

http://dx.doi.org/10.1016/j.sna.2017.07.032
0924-4247/

to reduce the pressure drop caused by the friction (which is high for diameters below 500 μm).

Authors in [8] describe the fabrication and test of a vaporizing liquid micro-thruster whose nozzle points in the direction normal to the chip plane. They built and tested two devices with different nozzle exit areas and tested them under different power conditions to characterize the thrust level per applied power.

The analytical modeling of a vaporizing liquid micro-thruster is shown in [9]. The thruster is the same presented in [8]. They focus on the formulation of the equations to calculate the power necessary to vaporize the liquid.

In [10] four micro resistojets were fabricated using MEMS technology and silicon wafers. The main differences among them are the type of nozzle (convergent or divergent) and the chamber volume (300 μm × 750 μm and 600 μm × 1500 μm).

Authors in [11] present the design and manufacturing techniques used to construct a micro resistojet consisting of an inlet portion, a heating section with one or more long channels, and a nozzle. Some tests were performed with three different devices: two with only one channel between the inlet and the nozzle with 10 μm and 5 μm wide, and other with 3 channels and 10 μm wide nozzle.

The devices found in literature often make use of complicated experimental setups including high performance data acquisition systems that might be incompatible with the limitations imposed by very small satellites such as CubeSats specially due to budgetary constrains since usually these satellites make extensive use of commercial-off-the-shelf components. Also, interfacing such a system to external components is always a challenge since these complex devices integrate fluidic, electrical, and mechanical characteristics into a very small device that also needs a reliable way of sensing the important parameters such as pressure and temperature.

In this paper, we present the design of VLMs with integrated heating and sensing capabilities that allow the easy operation of the microthrusters using standard commercial-off-the-shelf equipment. The devices are designed to meet the strict requirements of nano- and pico-satellites such as CubeSats and PocketQubes and operate using water as the propellant [12]. It has been shown that water is an interesting choice for micropropulsion applications as it can provide a very high Δv per volume of propellant when compared to other green propellants [13] making it very interesting for applications where orbital maneuvers are required. The heaters are made out of molybdenum which is a metal that can withstand very high temperatures (melting point 2693 °C) and can be patterned with standard dry or wet etching methods [14]. The resistivity of molybdenum is linearly proportional to the temperatures up to 700 °C allowing the design of heaters that also need precise temperature measurements as in the case of the VLMs. The structural design of the thrusters is based on previous work done in [15]. A special interface combining fluidic, electric, and mechanic connections has been developed to facilitate the operations of the thrusters right after dicing. The results of manufacturing and characterization of the VLMs are presented demonstrating the operations including feedback use of measurements of pressure and temperature. The devices have been manufactured with silicon wafers and tested under near-operational conditions in terms of pressure, mass flow, and power.

The remainder of this paper is organized as follows: Section 2 presents the background theory, Section 3 describes the designs of the thrusters, Section 5 shows the details of the experiments, the manufacturing process is described in Sections 4 and 6 presents the results of the tests, and finally Section 7 concludes the paper.

2. Background

2.1. Propulsion

The performance of micropropulsion systems can generally be analyzed using the same formulation as in normal sized systems. However, it is important to note that this formulation uses a set of assumptions that might not be applicable to micropropulsion systems as, for example, the assumption of negligible friction forces [16]. Thus, the following set of equations are used only to give insights into the ideal performance of such micropropulsion systems. In this case, two parameters are of major interest when analyzing the performance of the thruster: specific impulse and thrust. The thrust (F in Eq. (1)) is the force generated by the gas accelerated and expelled through the nozzle.

$$F = \dot{m}V_e + (p_e - p_a)A_e \tag{1}$$

where $\dot{m}$ is the mass flow rate, V_e is the exhaust velocity, p_e and p_a the exit and ambient pressures, and A_e is the exit area. The exhaust velocity can be calculated by (2) where M_e is the Mach number at the exit, k is the ratio of the specific heat at constant pressure and constant volume, T_1 is the chamber temperature, and R is the specific gas constant.

$$V_e = M_e\sqrt{kRT_1} \tag{2}$$

The mass flow rate can be written as a function of the chamber (stagnation) pressure and temperature (p_1 and T_1) and the area of the throat A_t:

$$\dot{m} = A_t p_1 k \frac{\sqrt{\left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}}}}{\sqrt{kRT_1}} \tag{3}$$

Eqs. (4)–(6) are used to calculate the Mach number, temperature, and pressure at the exit.

$$\frac{A_e}{A_t} = \left(\frac{k+1}{2}\right)^{-\frac{k+1}{2(k-1)}} M_e^{-1}\left(1 + \frac{k-1}{2}M_e^2\right)^{\frac{k+1}{2(k-1)}} \tag{4}$$

$$T_e = T_1\left(1 + \frac{(k-1)}{2}M_e{}^2\right)^{-1} \tag{5}$$

$$p_e = p_1\left(1 + \frac{(k-1)}{2}M_e{}^2\right)^{\frac{-k}{k-1}} \tag{6}$$

The specific impulse I_{sp} is a measure of efficiency regarding the consumption of propellant.

$$I_{sp} = \frac{F}{\dot{m}g} \tag{7}$$

where $g = 9.80665\ \mathrm{m\,s^{-2}}$ is the gravitational acceleration on Earth at sea level. Although the unit is given in seconds, it does not represent a measure of time but a measure of thrust per unit weight of propellant and it should be as high as possible for best propellant consumption efficiency.

Eqs. (1)–(7) are used to estimate the performance of the thrusters given the conditions of the experiments and the mechanical characterization of the devices.

2.2. Temperature dependent resistivity

The resistance of the heaters depends on the temperature and might be approximated by the following linear relation:

$$\alpha = \frac{R - R_0}{R_0(T - T_0)} \tag{8}$$

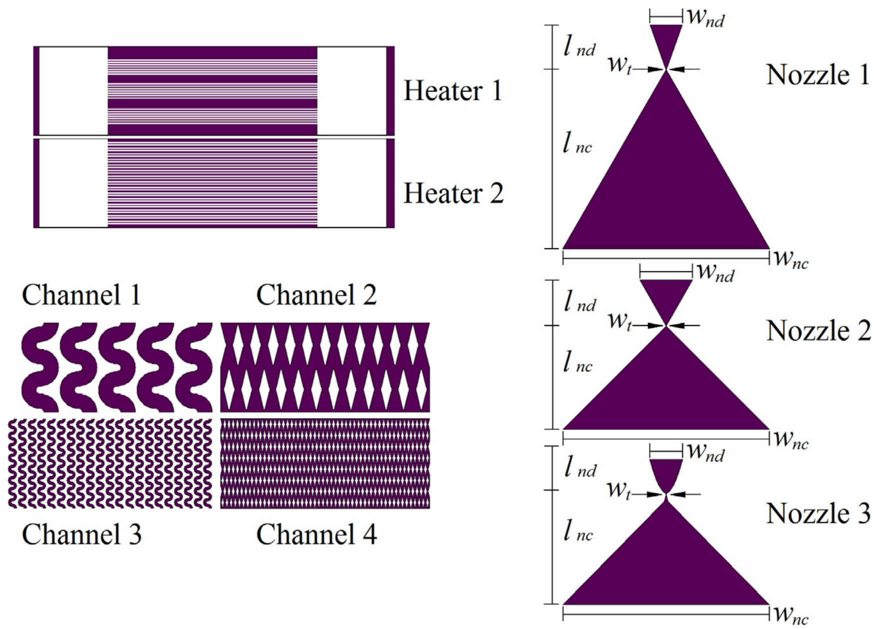

Fig. 1. Masks used in the manufacturing of the thrusters. The indexes *nc* and *nd* stand for the convergent and divergent parts of the nozzle respectively and the index *t* stands for the throat part of the nozzle. Dimensions are shown in Table 5.

where R is the resistance, T is the current temperature, and R_0 is the resistance measured at temperature T_0. The value of α can be characterized for each device and later used to estimate the temperature of the heaters. As the devices are very small and made of silicon which is good thermal conductor, a zero gradient of temperature might be assumed for the device. Therefore we can estimate that the temperature of the whole device is the same as the one estimated for the heaters. This approach has also been used to characterize similar devices made of molybdenum [14].

3. Design description

The microresistojets are composed of three main parts: a nozzle, a vaporizing chamber, and a heater. Each of these parts have different designs [17]. There are three types of nozzle named long nozzle, wide nozzle, and bell nozzle indicated in Fig. 1 as Nozzle 1, Nozzle 2 and Nozzle 3. The geometry and the area ratio (i.e. the ratio between the exit area of the nozzle and the throat area) are what differentiates them; the long nozzle has an area ratio of approximately 11, the wide nozzle 17, and the bell nozzle 11. For the vaporizing chamber there are four designs: diamond pillars that can be large or small with a total surface area per module of 7.09×10^{-6} m^2 and 2.27×10^{-5} m^2 shown in Fig. 1 as Channel 2 and Channel 4 respectively, and serpentine channels that can be large or small with a total surface area per module of 5.40×10^{-6} m^2 and 5.16×10^{-6} m^2 shown in Fig. 1 as Channel 1 and Channel 3 respectively. Finally, there are two types of heaters. One heater type containing 21 lines divided into three sets of 7 lines and the other containing 30 lines divided into sets of 2 lines; in both configurations the lines are 12 μm wide and 3000 μm long. Considering a sheet resistance of around $2\,\Omega\,\square^{-1}$ [14] the total resistance of each heater should be 3.40 Ω and 2.38 Ω respectively according to the following equation:

$$R = R_{sh}\frac{L}{W} \tag{9}$$

where R is the resistance, R_{sh} is the sheet resistance, L is the length of the resistance, and W is the width of the resistance.

The mask used during manufacturing containing all the needed features is presented in Fig. 1. The channels and the heaters are made in a modular manner such that a successive exposure of the slots is needed to make the complete chamber. Each module is designed to produce 1 W of power given a voltage of 5 V. In this case, seven modules are used in order to produce 7 W of power with 5 V which is the middle between maximum and minimum power needed for the thrusters as we will see in the next sub-section. The details of the channels are shown in Fig. 2. The dimensions are expected to be slightly different in the manufactured devices due to isotropic etching of the walls that erode approximately by 20 μm.

3.1. Propellant selection

Microresistojets can be designed to work with a variety of propellants in any state (gaseous, liquid, or solid) and the VLMs implicitly work with those that are liquid. Considering the range of pressure and temperature required by CubeSats and PocketQubes, the number of choices for the propellant is significantly reduced and considering the safety of the substance we can reduce it further. In a previous study [13], the authors have compared many substances that are applicable to microresistojets and the 10 most suitable propellants are listed in Table 1 with the respective performance values. The performance is evaluated for chamber pressures in the range of 2–5 bar, throat area of 5×10^{-9} m^2, and nozzle expansion ratio equal to 11.

Considering the fourth parameter, Δv per volume of propellant, water is the best candidate meaning that it can accelerate the spacecraft to higher velocities using the same volume of propellant. This is due to the fact that water is the most dense among all the candidates in conditions considered. If we consider the specific impulse, water has one of the highest values together with ammonia for the conditions analyzed. However, ammonia in its pure form is highly toxic which might not be suitable for use in CubeSats and PocketQubes. On the other hand, the power needed to vaporize water is the highest due to its high enthalpy of vaporization but it is still within reasonable range that can be provided by the electric power system of CubeSats and PocketQubes.

Therefore, water is the most suitable candidate for use in the VLMs as it outperforms the other candidates analyzed in terms of specific impulse and Δv which compensates the drawback of the power consumption. Thus, it has been selected as the propellant

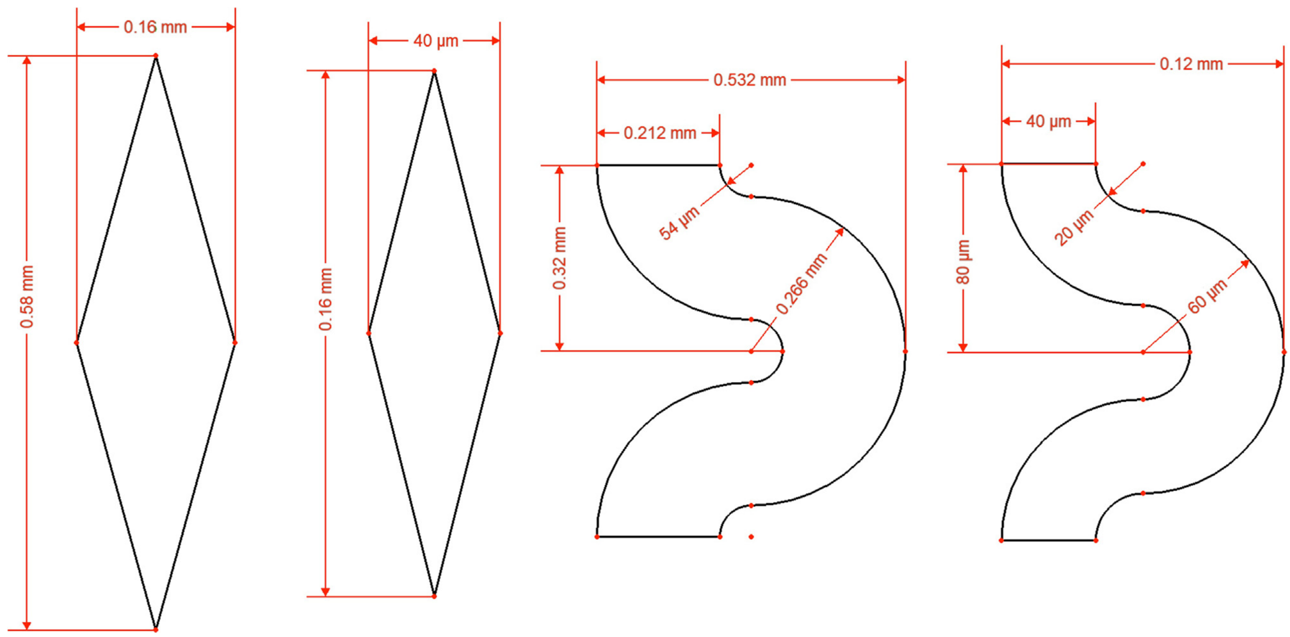

Fig. 2. Details of the design parameters of the channels. From left to right: large diamonds, small diamonds, large serpentine, and small serpentine.

Table 1
Comparison of different propellants that are suitable for use in VLMs [13].

Propellant	T (K)	F (mN)	I_{sp} (s)	$\Delta v/V$ ($m\,s^{-1}\,mL^{-1}$)	P (W)
Acetone	360–550	1.8–4.5	66.7–82.5	0.14–0.18	1.8–5.4
Ammonia	300–550	1.7–4.2	98.8–133.7	0.16–0.22	0.1–2.1
Butane	300–550	1.8–4.6	61.6–83.5	0.10–0.13	1.2–5.4
Cyclopropane	300–550	1.8–4.4	67.7–91.7	0.12–0.16	0.1–3.5
Decane	500–550	1.9–4.7	53.0–55.6	0.11–0.11	2.9–7.9
Ethanol	370–550	1.8–4.5	74.8–91.2	0.16–0.20	2.8–7.4
Isobutane	300–550	1.8–4.5	59.5–80.6	0.09–0.12	0.1–5.3
Methanol	360–550	1.7–4.3	83.2–102.8	0.18–0.22	2.7–7.1
Propene	300–550	1.8–4.5	69.5–94.1	0.10–0.13	0.1–2.6
Water	400–550	1.7–4.2	110.3–129.3	0.30–0.35	4.0–10.2

for use in the VLMs for the tests of operational characterization as well as the propellant for a future in-flight demonstration.

3.2. Performance parameters

With the design parameters of the nozzles, one can estimate the values of thrust and specific impulse given the conditions in the chamber using the equations introduced in Section 2.1. Assuming that the system is in the saturation point due to the boiling, the temperature in the chamber will vary in the range from 100 to 150 °C with the pressure in the range from 1 to 5 bar. Note, however, that it is still possible to heat the vapor to a higher temperature by providing more power to the heaters. With these conditions the thrust and specific impulse are calculated using the equations presented in Section 2 using the saturation temperature and pressure. The thrust and specific impulse for the long nozzle and the bell nozzle are the same as they have the same area ratio, respectively $F = 0.75$–3.79 mN and $I_{sp} = 105$–113 s. The wide nozzle is expected to have a slightly higher thrust $F = 0.77$–3.86 mN and specific impulse $I_{sp} = 107$–115 s. As all nozzles have the same throat area, the mass flow rate is the same for all of them in the range $\dot{m} = 0.73$–$3.42\ mg\,s^{-1}$. The power P necessary to heat-up the water from room temperature of about 24 °C to boiling point and vaporize it is calculated according to the following equation:

$$P = \dot{m}\Delta H \tag{10}$$

where $\Delta H = H_V - H_L$, H_V is the enthalpy of water vapor at boiling temperature, and H_L is the enthalpy of liquid water at room temperature. For the given mass flow rates, the power lies in the range $P = 1.87$–9.01 W.

4. Manufacturing

The starting material is a 100 mm double side polished silicon wafer with thickness of 300 µm. A layer of 500 nm of LPCVD (low pressure chemical vapor deposition) silicon nitride is deposited on the wafer to isolate the substrate from the heaters. Then, a layer of 200 nm of molybdenum is deposited on the front side of the wafer by sputtering. To form a hard mask for patterning Mo, a layer of PECVD TEOS (Plasma-enhanced chemical vapor deposition tetraethoxysilane) is deposited. The mask for the heaters is made with photoresist and the patterning is done first for TEOS (hard mask) by wet etching with buffered hydrofluoric acid (BHF) for approximately 2 min. Mo is etched with aluminum etch at 35 °C for approximately 26 s. The complete diagram with all the steps is shown if Fig. 4 and a schematic cross-section of the device is shown in Fig. 5.

After stripping off the photoresist with plasma cleaning and removing TEOS with BHF, a layer of 5 µm of silicon dioxide is deposited on both sides to form the hard mask of the cavities (chamber and inlet hole). Photoresist is used to form the soft mask for SiO_2 which is etched with plasma etching. The layer of silicon nitride is also etched in this step with a different recipe. The cavities of

Table 2
Codes used to identify the different thrusters.

Nozzle	L – Long	W – Wide	B – Bell	
Chamber	d – Diamond small	D – Diamond large	s – Serpentine small	S – Serpentine large
Heater	1–21 lines	2–30 lines		

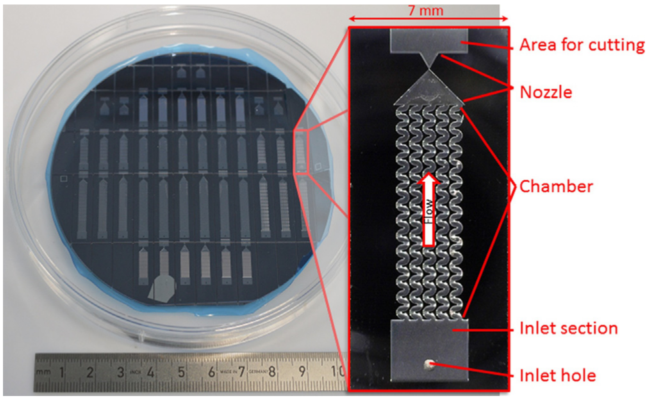

Fig. 3. In total each processed wafer yields 36 different devices plus 6 nozzle-only thrusters. In the right, one of the thrusters installed in the interface.

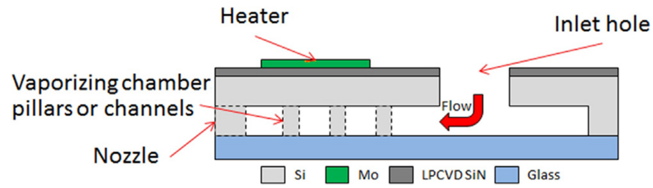

Fig. 5. Schematic cross-section of the thrusters. The chamber can be either with diamond pillars or serpentine channels.

the chamber are etched in silicon using a combination of isotropic and anisotropic deep reactive ion etching (DRIE). The isotropic step is done in order to make sure that the narrow channels are successfully opened. The fluid inlets are isotropically etched with DRIE from the heater side after the deposition of a layer of silicon dioxide on the chamber side.

Following the cleaning and removal of the hard masks, the silicon wafer is bonded to a glass wafer with anodic bonding at 400 °C and 1000 V. The glass is intended to provide a good visualization of the flow inside the thruster. This helps to understand the dynamics of the two-phase flow inside the chamber. The last step of manufacturing is dicing. Then the thrusters are ready for tests. A sample diced wafer and a thruster are shown in Fig. 3. The flow is from the bottom to the top: the propellant enters by the inlet hole, passes by the chamber where it is vaporized and is expelled to the environment by the nozzle. The area after the nozzle is added in order to have a safe path for the cutting blade during dicing; as show in Section 3 the nozzles can have different lengths. Therefore this area is to make sure that the shorter nozzles are open.

5. Experiment description

Two wafers have been processed using the steps described above. The first one was made without the inlet holes while the second was completely processed. The first wafer is intended to be used for electrical and mechanical tests only and was also used to first assess the bonding process. For these applications, the inlet hole is not needed. A single wafer yields 42 different devices that combine the different options described in Section 3. We selected ten thrusters for the mechanical and electrical experiments and four of them went through a further step of characterization to validate the devices in near-operational conditions. The complete list of thrusters is given in Table 3. Two devices (number 6 and 8) were found blocked, i.e. no flow can go from the inlet to the nozzle. This is probably due to incomplete etching or a defect caused by particles present during the manufacturing.

The thrusters received an identifier that describes the different design options. The code is in the form ‘XX-ABx-nn’ where ‘XX’ is the wafer number (00 or 01), ‘A’ is the type of nozzle (L – long, W – Wide, B – bell), ‘B’ is the type of chamber (d – diamond small, ‘D’

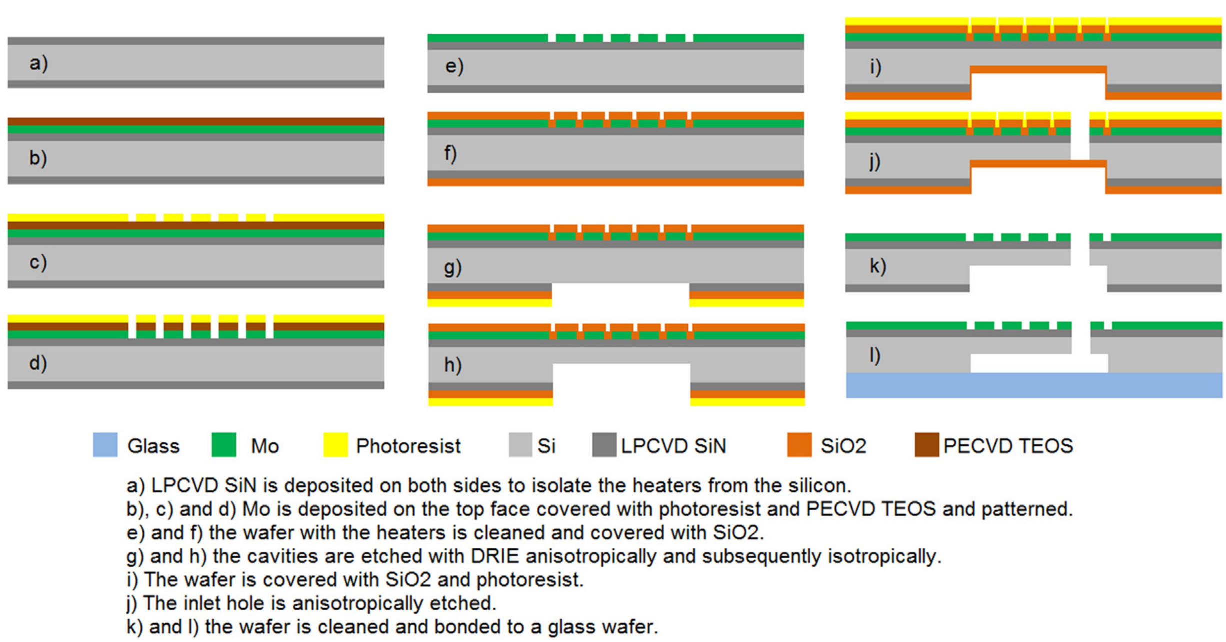

Fig. 4. Process flow diagram showing the steps of the manufacturing.

Table 3
List of selected thrusters. Test 1 is the mechanical test, test 2 is the electrical test, and test 3 is the operational test. The codes are presented according to the description given in Table 2.

Thruster	Code	Test 1	Test 2	Test 3	Detail
1	00-LD1-01	×	×		No inlet
2	00-Ld1-01	×	×		No inlet
3	00-WD2-01	×	×		No inlet
4	00-Bd2-01	×	×		No inlet
5	01-LS1-01	×	×	×	
6	01-BD1-01	×	×		Nozzle blocked
7	01-BS2-01	×	×	×	
8	01-WS2-01	×	×		Nozzle blocked
9	01-Ld1-01	×	×	×	
10	01-WD2-01	×	×	×	
11	01-Ws1-01	×			
12	01-Bs2-01	×			

Table 4
List of equipment used in the tests with the respective ranges and accuracies.

Equipment	Function	Output
ES 030-10	Power supply	0–30 V ± 5 mV/0–10 A ± 6mA
NE-1000	Pump – 2.5 mL syringe	2.5595×10^{-3}–186.15 mL h^{-1}
A35sc	Thermal camera	−40 to 160 ± 5 °C
MS5837-30BA	Pressure/temperature sensor	0–30 ± 0.1 bar/−20 to 85 ± 4 °C

– diamond large, 's' – serpentine small, 'S' – serpentine large), 'x' is the type of heater (1–21 lines, 2–30 lines), and 'nn' is the thruster number in case of repetition.

The test setup for the measurements is depicted in Fig. 6. It comprises a computer that controls the power supply for the heaters and the syringe pump used to pump water inside the thrusters at the desired flow rates. A list is presented in Table 4 with the ranges for each component and the respective accuracy. The biggest source of inaccuracies is related to the mechanical movement of the pump that causes small variations in the flow rate that can be seen in the pressure measurements as reported by other authors [18–20]. The data of temperature and pressure from the interface of the thruster is sent to the computer which also measures the voltage and current levels of the power supply. For the electrical characterization of the resistances, no water is injected into the thruster.

A special interface, shown in Fig. 7, has been designed made out of Teflon and aluminum to allow the easy operation of the thrusters. This interface provides the means for electrical connection to the heaters and also a leak-free connection for the propellant feeding line. Also, a digital sensor measures the pressure and temperature of the liquid being injected into the thruster. The sensor measures pressures in the range 0–30 ± 0.1 bar and temperatures in the range from −20 to 85 ± 4 °C.

5.1. Mechanical characterization

All the devices have been subject to a mechanical characterization in order to evaluate the dimensions of the features and compare them with the designed values. Table 5 presents the design parameters for the structures. The values d_1 and d_2 refer to the dimensions of the channels. In the case of diamond pillars, these are the sizes of the diamonds as indicated in Fig. 2; in the case of serpentine channels, these are the inner and outer radii of the semi-circles. The surface roughness is also characterized in order to evaluate the results of the last step of the etching process (isotropic etching). The effect of the surface roughness is an important parameter that affects the performance of microthrusters and has been reported elsewhere [21,22].

Table 5
Parameters used in the design of the thrusters. All dimensions are in μm. Type descriptions can be found in Table 2.

Type	w_{nd}	l_{nd}	w_{nc}	l_{nc}	w_t
L	500	645	3000	2600	45
W	780	660	3000	1500	45
B	500	500	3000	1600	45
		d_1		d_2	
d		160		40	
D		580		160	
s		60		20	
S		266		54	

5.2. Electrical characterization

The devices were subject to a resistance test in order to characterize the temperature resistance coefficient α. The test consists in applying a constant voltage to the heaters, measuring the current passing through them, and recording images with the thermal camera. Fig. 8 shows a sequence of images taken with the infra-red camera from the nozzle exit plane. The nozzle face of the thruster is in the middle of the image where we see the highest temperatures. Thus, the flow vector is normal the plane of the picture. This approach provides an acceptable estimation of the temperature, given the assumption that the whole thruster is at the same temperature.

The images of the thermal camera are then used to estimate the average temperature of the thruster over time. This average temperature is used to estimate α for each thruster. The test starts with applying very low power to the heaters in order to measure the initial resistance. Then the power is increased in two steps applying first 5 V till the temperature stabilizes at a certain value (i.e. reaches steady state) and 7 V till it reaches steady state. Based on the measurements for each device, the value of α is estimated using (8). In total, 10 devices have been calibrated with this method.

5.3. Operational characterization

In the last step of characterization, four devices have been tested in order to validate the devices in near-operational conditions. The test consists of injecting water into the thruster and applying power to the heaters to vaporize the water. The data from the sensors, i.e. pressure and temperature, and from the power supply is collected and also used during the tests as feedback information for manual control of the input variables (flow rate and power).

Water is pumped into the thrusters using the pump with a glass syringe of 2.5 mL to provide a constant mass flow rate sufficient to achieve a pressure of 5 bar in the vaporizing chamber. Although it is a reliable way of controlling the thrusters, it is only applicable for experimental testing since in does not represent the system as in the real application which will consist of a pressurized tank that provides the propellants to the thruster. Also, as already mentioned the syringe pump causes low frequency pressure fluctuations in the flow. These fluctuations might affect the stability of the two-phase flow inside the chamber (i.e. droplets can be ejected during the peaks in pressure).

The manual control is done based on the visual behavior of the vaporization. The power and mass flow rate are manually set to such values that correspond to a complete vaporization of the water without spotting any droplets coming out of the nozzle. The avoidance of droplets is very important because the heaters, made of Mo, can oxidize very easily when in contact with water at high temperatures. A video showing one of the tests is available at [23]. Fig. 9 shows some snapshots at different moments of a test.

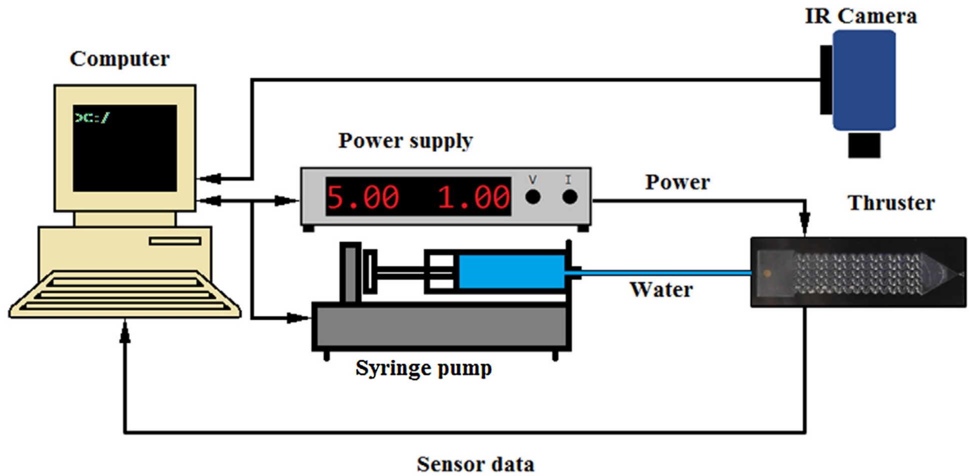

Fig. 6. Test configuration for electrical and operational characterization. The infra-red camera is only used during the electrical characterization.

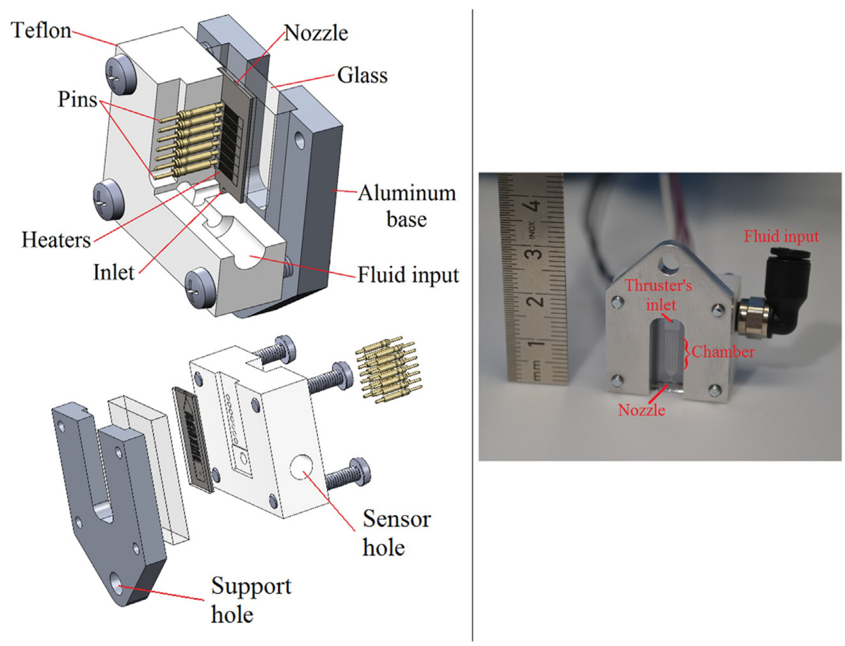

Fig. 7. Interface for easy operation of the thrusters. Top left corner: a cut showing the details of the pins and the fluidic channels; bottom left corner: an exploded view of the 3D model; right side: a photo of the interface. The pressure and temperature sensor are on the side opposite to the fluid input. The spring loaded pins connect the heaters to the power supply.

A steady state is achieved when the complete boiling is occurring at an arbitrary section of the chamber. At this operating state, the power and mass flow rate are adjusted to achieve a constant pressure in the chamber of approximately 5 bar.

6. Results and discussion

6.1. Mechanical characterization

The structural dimensions have been assessed as described in Section 5 with an optical microscope. In Table 6 the average measured values are shown and the designed values are in brackets. In general, the averages are slightly smaller than what they were designed. The heaters were also measured in order to compare with the designed values. On average each line of the heaters is around 11.3 μm.

The depth of the cavities has been measured with a Dektak Surface Profiler and on average they are 100 μm. With these measurements we can recalculate the performance parameters estimated in Section 3. As we see in Table 7, the mass flow and the thrust are reduced almost by half due to the reduction in the throat width while the specific impulse remains the same. As a consequence the power needed also decreases. Although the differences seen are acceptable, this indicates the need for more precision in the manufacturing to reduce differences between the designed and manufactured devices.

The characterization of the surface roughness was done with AFM (Atomic Force Microscopy) using a nTEGRA Aura AFM with NSG10 tip (10 nm tip radius) and a 100 μm closed-loop sample scanner. The measurement was done in an area of 40 μm × 40 μm of a sample that has not been completed in terms of manufacturing since an open device is needed for the measurement. As the last step of the manufacturing is the isotropic etching, we can assume that the roughness of all the surfaces (including the nozzle throat) is similar to the measured roughness of the sample. Table 8 shows

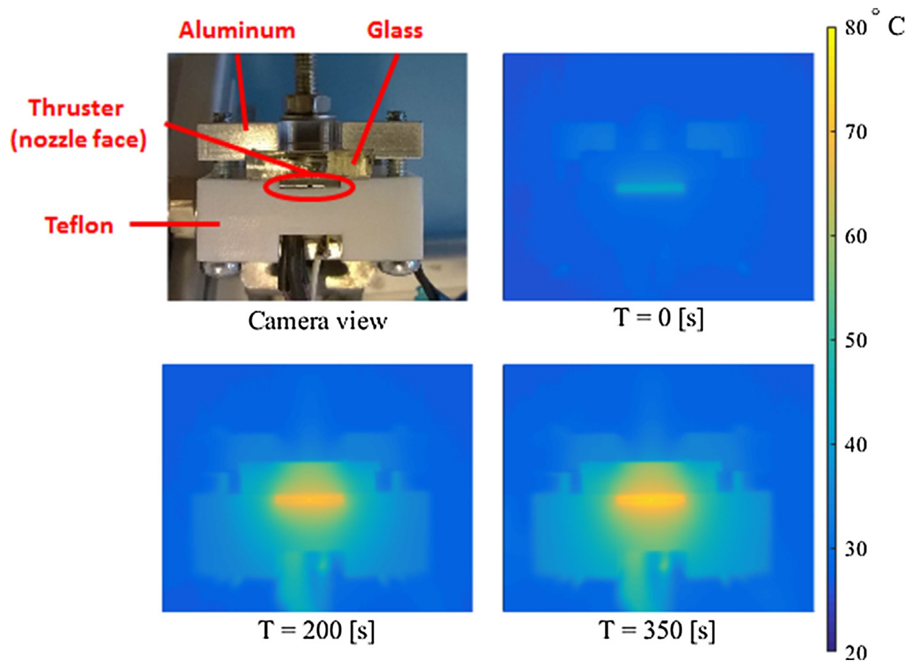

Fig. 8. Thermal images of different moments of the test.

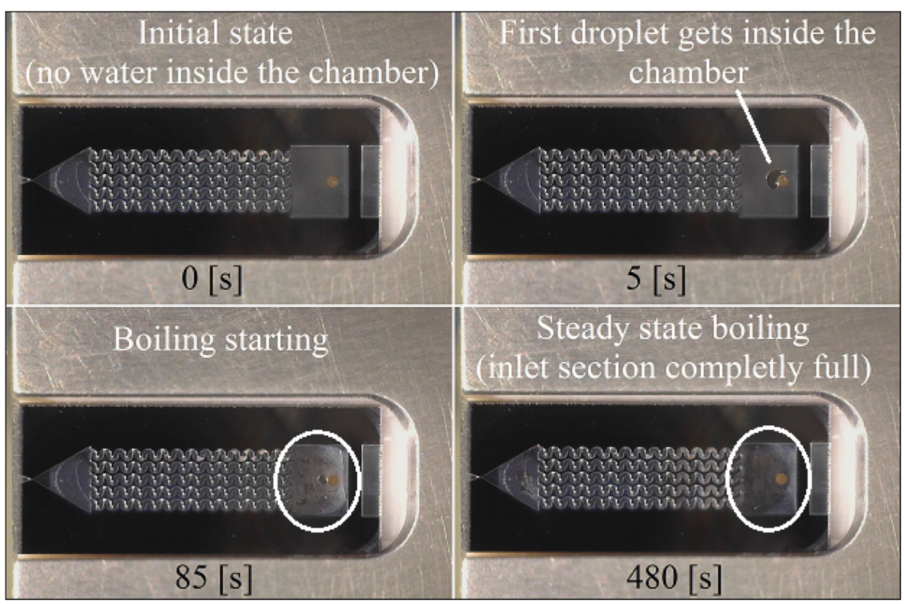

Fig. 9. The water is pumped very slowly in the beginning in order to avoid instabilities. Top left corner: water has not yet come inside the chamber; top right corner: water just getting in the chamber; bottom left corner: chamber partially full and boiling; bottom right corner: complete boiling during steady state.

Table 6
Average measured values of the dimensions of the thrusters in μm. Design values are in brackets.

Type	w_{nd}	l_{nd}	w_{nc}	l_{nc}	w_t
L	489.2 ± 2.7 (500)	626.0 ± 4.2 (645)	2979.0 ± 5.6 (3000)	2549.2 ± 22.9 (2600)	25.1 ± 3.5 (45)
W	777.7 ± 1.5 (780)	643.6 ± 2.9 (660)	2980.4 ± 4.9 (3000)	1489.5 ± 20.9 (1500)	26.0 ± 2.8 (45)
B	492.6 ± 4.5 (500)	486.9 ± 2.3 (500)	2983.3 ± 10.2 (3000)	1581.3 ± 13.0 (1600)	20.1 ± 3.2 (45)

	d_1	d_2
d	10.3 ± 0.3 (160)	2.8 ± 0.1 (40)
D	547.7 ± 4.9 (580)	144.3 ± 4.7 (160)
s	76.6 ± 0.1 (60)	8.2 ± 0.9 (20)
S	289.7 ± 7.2 (266)	39.9 ± 2.1 (54)

Table 7
Comparison between the design parameters (in brackets) and the ones calculated with the measured values.

Nozzle	$\dot{m}$ (mg s^{-1})	F (mN)	I_{sp} (s)	P (W)
L	0.41–1.91 (0.73–3.42)	0.43–2.16 (0.75–3.79)	107.34–115.60 (105.22–112.89)	1.04–5.03 (1.87–9.01)
W	0.42–1.97 (0.73–3.42)	0.45–2.27 (0.77–3.86)	108.34–117.22 (106.98–115.09)	1.08–5.20 (1.87–9.01)
B	0.33–1.53 (0.73–3.42)	0.35–1.75 (0.75–3.79)	107.94–116.51 (105.22–112.89)	0.84–4.03 (1.87–9.01)

Table 8
Surface roughness statistical values.

Property	Value
Minimum	−338.671 nm
Maximum	391.408 nm
Average value	11.412 nm
Median	12.999 nm
R_a (S_a)	65.223 nm
R_{RMS} (S_q)	81.803 nm
R_{RMS} (grain-wise)	81.803 nm
Skew	−0.152
Kurtosis	0.1514
Surface area	1.67834×10^{-9} m^2
Projected area	1.60157×10^{-9} m^2
Variation	438.125 μm^2
Entropy	−14.903
Entropy deficit	0.0029304
Inclination θ	0.03°
Inclination ϕ	165.26°

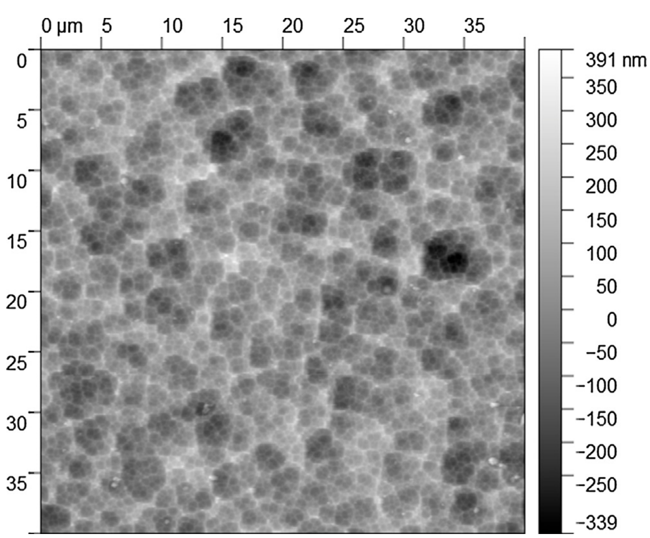

Fig. 10. Surface roughness of a sample.

the average values for the measurement. Fig. 10 shows the image taken with AFM.

6.2. Electrical characterization

In Fig. 11 the measurements recorded with the camera are plotted against the resistance measured for four devices. These measurements (done with 10 devices) are used to estimate the value of α as shown in Table 9. The average value is $\bar{\alpha} = 1.09 \times 10^{-3}\,{}^{\circ}\mathrm{C}^{-1}$ and standard deviation $\sigma_\alpha = 1.79 \times 10^{-4}$ and can be used for other devices that have been manufactured together. An accurate measurement of the initial resistance at room temperature is very important to avoid discrepancies in the estimation of temperature. Fig. 12 shows the comparison between the measurements with the thermal camera and the estimation based on (8). The noise seen in the plots comes from the experimental setup that was built using standard equipment with limited precision. This is useful in order mimic the conditions in which the thruster will eventually operate, i.e. in a very small satellite with limited electronics in a harsh environment (i.e. space). As we can see, without considering the noise, the estimation matches very well with the measurements.

Table 9
Values of α for all the thrusters tested.

Thruster	Code	α (°C^{-1})
1	00-LD1-01	1.30×10^{-3}
2	00-Ld1-01	1.33×10^{-3}
3	00-WD2-01	1.28×10^{-3}
4	00-Bd2-01	8.76×10^{-4}
5	01-LS1-01	1.22×10^{-3}
6	01-BD1-01	1.09×10^{-3}
7	01-BS2-01	9.73×10^{-4}
8	01-WS2-01	8.61×10^{-4}
9	01-Ld1-01	9.55×10^{-4}
10	01-WD2-01	1.01×10^{-3}
Average		1.09×10^{-3}
Standard deviation		1.79×10^{-4}

6.3. Operational characterization

As mentioned earlier, four thrusters have been tested with water under near-operational conditions but with ambient pressure equal to atmospheric pressure. This high ambient pressure degrades the thrust efficiency but the purpose is to demonstrate the operation of the thrusters.

Water is pumped inside the thruster (very slowly in the beginning) and, after it reaches the inlet section, the power is increased to start the vaporization. When this happens the pressure increases and the power is manually controlled to allow full vaporization (visually) of the propellant. Then, the mass flow rate and the power are kept constant in order to maintain the pressure in the chamber at approximately 5 bar (as mentioned before the control is done manually and with visual feedback). In Fig. 13 the values of pressure, power, and mass flow rate during the steady state part of the experiment are plotted. We can see that the mass flow rate is not the same even though the nozzle throat area is not much different. The measured values for the throat width are 23.1, 16.5, 20.9 and 23.6 μm for thrusters 5, 7, 9 and 10 respectively. This fact might be attributed to the efficiency in the vaporization which is dependent on the shape of the microchannels. In Table 10 we can see the difference in energy per milli-gram of water used in the process. It is clearly seen that the one with the small diamonds (thruster 9) is the most efficient due to its larger surface area (it is 3.5, 3.6, and 2.9 times larger than thrusters 5, 7, and 10 respectively). We can also see in Fig. 13 the variation in the pressure that is caused by the syringe pump as described in Section 5. Although this effect is clearly visible in the pressure measurements, no instability (i.e. no droplets coming out of the nozzle) was spotted during the steady state.

Considering the applied mass flow rate, we can recalculate the power needed to heat-up and vaporize the water and subtract it from the measured power in order to estimate the losses to the structure and the environment which on average for all the tests is around $P_{loss} = 6.17$ W. This represents an efficiency in the energy use of around 23% on average.

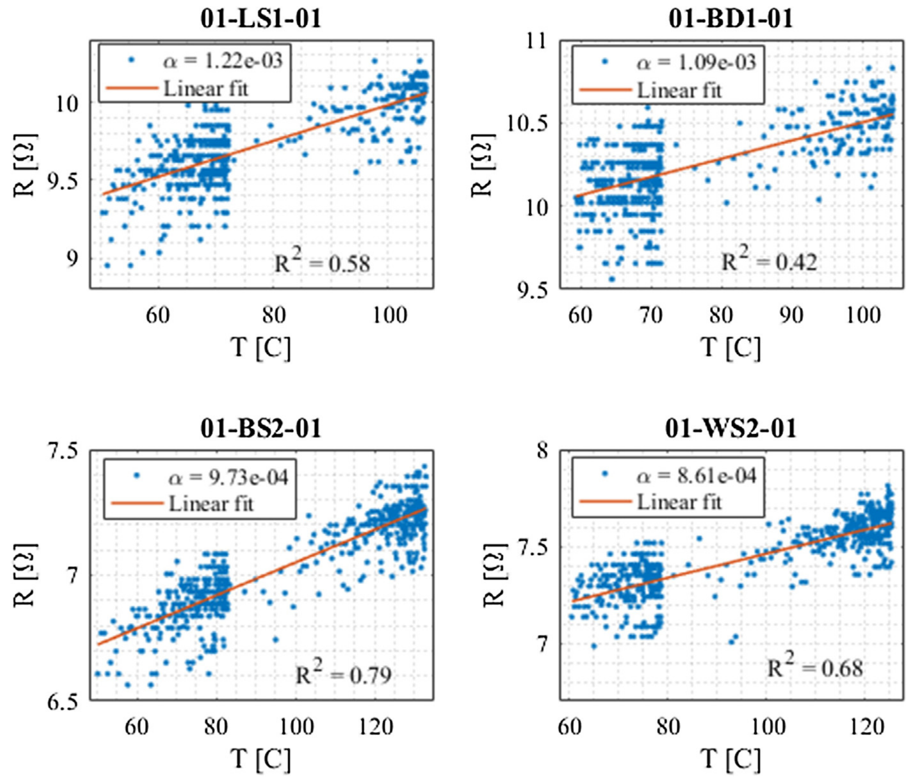

Fig. 11. Resistance of the heaters as a function of the temperature for 4 devices.

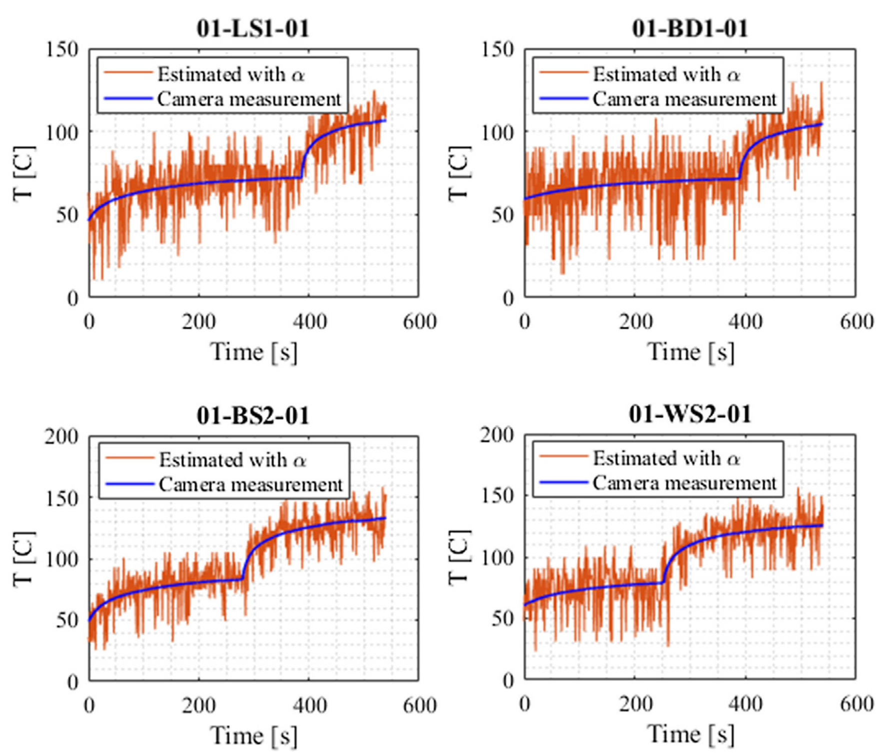

Fig. 12. Comparison between estimated and measured values of the temperature of four thrusters.

Fig. 14 shows the resistances measured during the steady state and the temperature estimated with these values. Given the pressure of water around 5 bar, the saturation temperature is around 151.83 °C. As the power is controlled such that full vaporization occurs, we can consider that this is the temperature that should be measured. This approach is valid for characterization, however in the real operations of the thrusters the power can be set higher to further increase the temperature of the vapor. We can see that the estimated temperature is slightly different for each thruster which

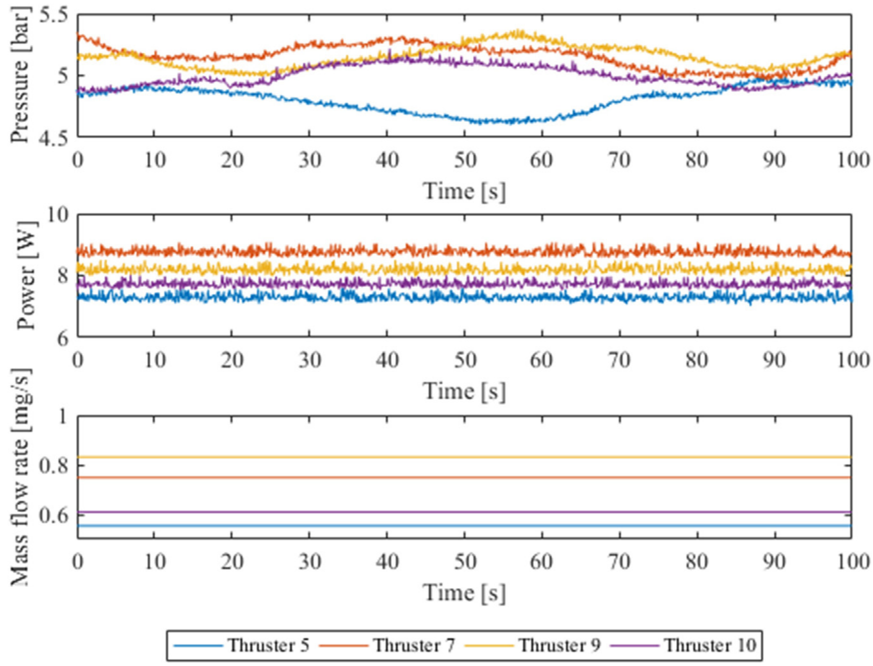

Fig. 13. Measured pressure, power, and mass flow rate for the tested devices. The low frequency variation in the pressure is caused by mechanical oscillations in the pump.

Table 10
Average values measured during steady state.

Thruster	P (bar)	$\dot{m}$ (mg s^{-1})	P_{meas} (W)	P_{calc} (W)	E (J mg^{-1})	Efficiency %
5	4.80	0.55	7.29	1.47	13.16	20
7	5.15	0.75	8.76	1.99	11.71	23
9	5.15	0.83	8.19	2.21	9.84	27
10	5.00	0.61	7.72	1.62	12.66	21

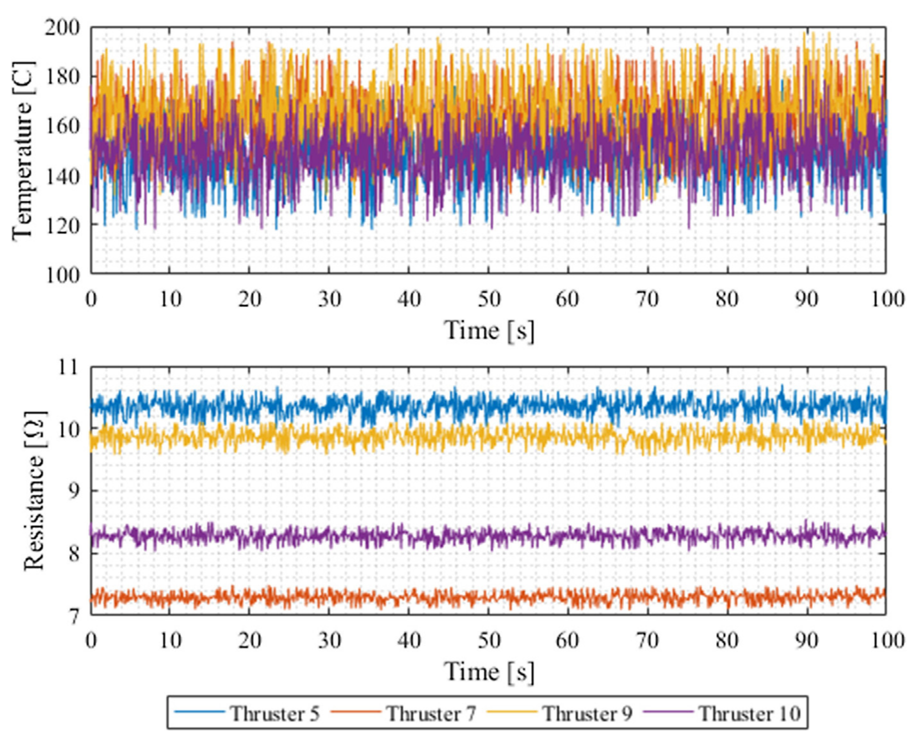

Fig. 14. Estimated temperature derived from the measured resistance for the tested devices.

Table 11
Comparison between the results of this work and other references. The values marked with * are calculated or simulated.

Reference	P (W)	$\dot{m}$ (mg/s)	F (mN)	I_{sp} (s)	p (bar)	T (K)	τ (mN/W)
[24]	n/a	2.33–8.33	2–6.5	65–105	1.0–2.6	454–574	n/a
[5]	4.01	1.0	0.634	31	n/a	n/a	0.16
[25]	n/a	2.08–16.6*	1–6*	48.9–36.9*	1.0–2.0	423–573	n/a
[6]	7.1–9.2	1.0	0.034–0.07	3.42–6.9	1.0	400–422	0.005–0.007
[7]	1.6–3.6	0.2–2.04	0.15–1.01	50–105	1.0	374–474	0.09–0.28
[9]	1–2.4	0.7	0.005–0.12	17.5*	n/a	n/a	0.01–0.05
[26]	10.8	8.8	0.46	5.33*	n/a	n/a	0.04
[10]	30	0.038	0.003	7.78*	n/a	n/a	0.0001
This work							
Thruster 5	7.29	0.55	0.67*	124.02*	4.80	423.03	0.09
Thruster 7	8.76	0.75	0.88*	119.80*	5.15	425.65	0.10
Thruster 9	8.19	0.83	0.98*	120.20*	5.15	425.65	0.12
Thruster 10	7.72	0.61	0.74*	123.72*	5.00	424.54	0.10

can be attributed to differences in the measurements of the initial resistances and initial temperature.

Table 11 presents a comparison of the expected performance in terms of thrust, specific impulse and power of the devices tested with water and the results of other references found in the literature. The last column shows the thrust to power ratio which is calculated dividing the thrust by the power. The level of thrust is in a range comparable to other devices but it needs to be experimentally measured. The power is the power needed to heat up not only the thruster and the propellant but also the complete interface which has not yet been optimized in terms of thermal isolation. Also, the tests were conducted in room conditions which means that a significant amount of power is lost to the environment. Therefore, the power consumption of an optimized system in vacuum is expected to be significantly lower than these values. However, looking at the parameter τ we can see that the (calculated) delivered thrust per unit power is already in a range among the highest values.

7. Conclusions

This paper presented the details of the design, manufacturing, and characterization of microresistojets with integrated heating and temperature measurement capabilities that operate by vaporizing water and accelerating the vapor with a convergent-divergent nozzle. In total, 12 devices have been assessed for their mechanical characteristics while 10 of those have undergone an electrical characterization to allow the estimation of temperature using measurements of resistance. Finally, four devices have been tested in near-operational conditions in order to validate the current thruster design and have a glimpse into the operational characteristics of such devices. We successfully demonstrate the operations of the devices using water as propellant and having online measurements of temperature that can be used for feedback control, for example. The results presented here serve to determine the fundamental operational and design characteristics of this kind of propulsion system and will be useful in future implementations.

The manufacturing process has been effective in the sense that most of the devices could be tested with water with the exception of two that have been found blocked. The use of a glass wafer to cover the cavities was very important since it provides a good visualization of the vaporization process and allows the visual control of the operation which would be much more difficult without it. Also, it provides valuable information for automatic control of temperature of the semi stochastic boiling process. However, this will not be necessary in the flight models where no flow visualization is required and a pure silicon wafer can be used.

The use of molybdenum heaters has proved to be a very effective design choice since it is very stable at the temperatures used and can also achieve very high temperatures up to 850 °C. However, the heaters need a protection, such as PECVD silicon oxide, to avoid oxidation at temperatures higher than 350 °C. The measurements of temperature were used to control the process and keep the vaporization stable showing the reliability of this technique. The modularity of the design allows the measurement of temperature at different positions in the chamber and will be investigated in the future.

The developed interface for testing the thrusters has proved to be very robust and easy to use. The thrusters can be tested right after dicing and they can be reconnected to the interface in a couple of minutes reducing testing time. It provides a good way of connecting the heaters to a power supply without the need of wire bonding and also a leak-free connection for the fluid inlet. The sensor included in the interface provides a measurement of pressure very close to the chamber such that it is safe to assume that the measured values are representative of the ones inside the chamber. This interface also facilitates the fast replacement of thrusters without damaging the devices.

Compared to other devices found in the literature, the tested thrusters have shown a performance close to the highest specially in terms of thrust to power ratio (τ) which is very interesting for nano- and pico-satellites that have limited capabilities in power generation. However, the values of thrust have been calculated based on the measurements of pressure, temperature, and mass flow. In order to validate these values, a direct measurement of the thrust has to be done.

Future research will be focused on the optimization of the channels in the heating chamber in order to improve the heat transfer and the power consumption. A control loop will also be designed to improve the operational performance and focused on the development of a propulsion system that can be used in nano- and pico-satellites. To this purpose, the integration of electronics for the power control of the heaters is currently under investigation. This will represent a large step towards a fully integrated device. The interface for the thrusters will also be improved focused on the reduction of size and thermal isolation. The manufacturing process will be improved by reducing the number of steps, specially in the etching of the cavities, and by including a protection in the heaters to reduce problems with oxidation in high temperatures. The measurements of initial resistance and temperature can be improved in favor of the electrical characterization as well as the measurement of resistance during operation to reduce noise in the estimated temperature. The calibration process will be improved by testing the heaters in a hot plate in vacuum. This is expected to increase the precision of the measurements since a more homogeneous distribution of temperature can be achieved. To this purpose, a new interface has to be developed to facilitate these measurements. More tests will be carried out in vacuum to reduce the heat losses and also using a thrust measurement bench to measure the actual thrust and specific impulse of the devices. Finally, the manufacturing process as well as the mask design will be improved to facilitate the fabrication by reducing process steps and also to reduce differences between the designed and the manufactured devices.

Acknowledgments

The present work was supported by CNPq, Conselho Nacional de Desenvolvimento Científico e Tecnológico, Brasil.

References

[1] J. Bouwmeester, J. Guo, Survey of worldwide pico- and nanosatellite missions, distributions and subsystem technology, Acta Astronaut. 67 (78) (2010) 854–862, http://dx.doi.org/10.1016/j.actaastro.2010.06.004.

[2] D. Selva, D. Krejci, A survey and assessment of the capabilities of cubesats for earth observation, Acta Astronaut. 74 (2012) 50–68, http://dx.doi.org/10.1016/j.actaastro.2011.12.014.

[3] C. Boshuizen, J. Mason, P. Klupar, S. Spanhake, Results from the planet labs flock constellation, in: 28th Annual AIAA/USU Conference on Small Satellites, 2014.

[4] K. Lemmer, Propulsion for cubesats, Acta Astronaut. (2017), http://dx.doi.org/10.1016/j.actaastro.2017.01.048, in press, accepted manuscript.

[5] K.H. Cheah, K.-S. Low, Fabrication and performance evaluation of a high temperature co-fired ceramic vaporizing liquid microthruster, J. Micromech. Microeng. 25 (1) (2015) 015013, http://dx.doi.org/10.1088/0960-1317/25/1/015013.

[6] K. Karthikeyan, S. Chou, L. Khoong, Y. Tan, C. Lu, W. Yang, Low temperature co-fired ceramic vaporizing liquid microthruster for microspace-craft applications, Appl. Energy 97 (0) (2012) 577–583, http://dx.doi.org/10.1016/j.apenergy.2011.11.078, Energy Solutions for a Sustainable World – Proceedings of the Third International Conference on Applied Energy, May 16–18, 2011 – Perugia, Italy.

[7] P. Kundu, T.K. Bhattacharyya, S. Das, Design, fabrication and performance evaluation of a vaporizing liquid microthruster, J. Micromech. Microeng. 22 (2) (2012) 025016, http://dx.doi.org/10.1088/0960-1317/22/2/025016.

[8] D. Maurya, S. Das, S. Lahiri, An analytical model of a silicon mems vaporizing liquid microthruster and some experimental studies, Sens. Actuators A: Phys. 122 (1) (2005) 159–166, http://dx.doi.org/10.1016/j.sna.2005.04.020, sSSAMW 04 Special Section of the Micromechanics Section of Sensors and Actuators based on contributions revised from the Technical Digest of the 2004 Solid-State Sensor, Actuator and Microsystems Workshop.

[9] D.K. Maurya, S. Das, S.K. Lahiri, Silicon mems vaporizing liquid microthruster with internal microheater, J. Micromech. Microeng. 15 (5) (2005) 966, http://dx.doi.org/10.1088/0960-1317/15/5/010.

[10] X. Ye, F. Tang, H. Ding, Z. Zhou, Study of a vaporizing water micro-thruster, Sens. Actuators A: Phys. 89 (12) (2001) 159–165, http://dx.doi.org/10.1016/S0924-4247(00)00540-9, Special Issue: Micromechanics Section of Sensors and Actuators, based on contributions revised from the Technical Digest of the Thirteenth IEEE International Workshop on Micro Electro Mechanical Systems (MEMS-2000).

[11] M. Mihailovic, T. Mathew, J. Creemer, B. Zandbergen, P. Sarro, MEMS silicon-based resistojet micro-thruster for attitude control of nano-satellites, in: 2011 16th International Solid-State Sensors, Actuators and Microsystems Conference (TRANS-DUCERS), 2011, pp. 262–265, http://dx.doi.org/10.1109/TRANSDUCERS.2011.5969432.

[12] A. Cervone, V. Pallichadath, M.A.C. Silva, D. Guerrieri, S. Silvestrini, S. Mestry, D. Maxence, B. Zandbergen, H.V. Zeijl, MEMS water micro-resistojets for CubeSats and PocketQubes, in: MPCS 2017 – International Workshop on Micropropulsion and CubeSats, 2017.

[13] D.C. Guerrieri, M.A.C. Silva, A. Cervone, E. Gill, Selection and characterization of green propellants for micro-resistojets, ASME J. Heat Transfer 139 (10) (2017) 9, http://dx.doi.org/10.1115/1.4036619.

[14] L. Mele, F. Santagata, E. Iervolino, M. Mihailovic, T. Rossi, A. Tran, H. Schellevis, J. Creemer, P. Sarro, A molybdenum MEMS microhotplate for high-temperature operation, Sens. Actuators A: Phys. 188 (2012) 173–180, http://dx.doi.org/10.1016/j.sna.2011.11.023, Selected papers from the 16th International Conference on Solid-State Sensors, Actuators and Microsystems.

[15] R. Poyck, I. Krusharev, Q. Bellini, B. Zandbergen, A. Cervone, A water-fed micro-resistojet for the delffi formation flying mission, in: International Astronautical Congress, 2014.

[16] G.P. Sutton, O. Biblarz, Rocket Propulsion Elements, 8th ed., John Wiley & Sons, 2010.

[17] A. Cervone, B. Zandbergen, D.C. Guerrieri, M. Athayde Costa e Silva, I. Krusharev, H. van Zeijl, Green micro-resistojet research at Delft University of Technology: new options for cubesat propulsion, CEAS Space J. Online First (2016) 1–15, http://dx.doi.org/10.1007/s12567-016-0135-3.

[18] W. Zeng, I. Jacobi, D.J. Beck, S. Li, H.A. Stone, Characterization of syringe-pump-driven induced pressure fluctuations in elastic microchannels, Lab Chip 15 (2015) 1110–1115, http://dx.doi.org/10.1039/C4LC01347F.

[19] W. Zeng, I. Jacobi, S. Li, H.A. Stone, Variation in polydispersity in pump- and pressure-driven micro-droplet generators, J. Micromech. Microeng. 25 (11) (2015) 115015, http://dx.doi.org/10.1088/0960-1317/25/11/115015.

[20] P.M. Korczyk, O. Cybulski, S. Makulska, P. Garstecki, Effects of unsteadiness of the rates of flow on the dynamics of formation of droplets in micro fluidic systems, Lab Chip 11 (2011) 173–175, http://dx.doi.org/10.1039/C0LC00088D.

[21] F. La Torre, S. Kenjeres, C.R. Kleijn, J.-L.P. Moerel, Effects of wavy surface roughness on the performance of micronozzles, J. Propuls. Power 26 (4) (2010) 655–662, http://dx.doi.org/10.2514/1.44828.

[22] Y. Cai, Z. Liu, Z. Shi, Effects of dimensional size and surface roughness on service performance for a micro laval nozzle, J. Micromech. Microeng. 27 (5) (2017) 055001, http://dx.doi.org/10.1088/1361-6439/aa6552.

[23] M.A.C. Silva, Vaporizing Liquid Microthruster, 2017, accessed (01.03.17).

[24] J. Cen, J. Xu, Performance evaluation and flow visualization of a MEMS based vaporizing liquid micro-thruster, Acta Astronaut. 67 (34) (2010) 468–482, http://dx.doi.org/10.1016/j.actaastro.2010.04.009.

[25] C.-C. Chen, C.-W. Liu, H.-C. Kan, L.-H. Hu, G.-S. Chang, M.-C. Cheng, B.-T. Dai, Simulation and experiment research on vaporizing liquid micro-thruster, Sens. Actuators A: Phys. 157 (1) (2010) 140–149, http://dx.doi.org/10.1016/j.sna.2009.10.025.

[26] E. Mukerjee, A. Wallace, K. Yan, D. Howard, R. Smith, S. Collins, Vaporizing liquid microthruster, Sens. Actuators A: Phys. 83 (13) (2000) 231–236, http://dx.doi.org/10.1016/S0924-4247(99)00389-1.

Biographies

Marsil A.C. Silva graduated from the Pontifical Catholic University of Parana with a degree in Mechatronics Engineering (Control and Automation) in 2012. In 2014 he earned a Master's degree in Mechanical and Aeronautical Engineering from Instituto Tecnológico de Aeronáutica in Brazil. He started a PhD in 2015at the Faculty of Aerospace Engineering of TU Delft and is working on thrust control for micropropulsion systems. His research interests are on MEMS, control systems, system identification, micropropulsion, artificial intelligence and optimization.

Daduí C. Guerrieri graduated from Universidade Estadual do Rio de Janeiro (UERJ) with a Bachelor degree in Mechanical Engineering in 2011. In 2014, he received a Master of Science in Mechanical Engineering from Universidade Federal do Rio de Janeiro (UFRJ). Since 2012, he is a lecturer from Centro Federal de Tecnologia do Rio de Janeiro (CEFET-RJ). Since 2014, he is a PhD candidate at the department of Space Engineering at TU Delft working on the topic of micro-resistojets. In addition, he is part of the microfluidics and microsystems committee at the Associação Brasileira de Engenharia e Ciências Mecânicas (ABCM). His research interests include MEMS, microfluidics, thermal science, heat transfer, and propulsion systems.

Henk van Zeijl received the B.S. degree in physics from Polytechnical Institute, Rijswijk, The Netherlands, in 1981. He received the Ph.D. degree from the Delft University of Technology, Delft, The Netherlands, in 2005, based on research on bipolar transistors with self aligned emitter-base metallization and back-wafer-aligned collector contacts. He joined the Interuniversity Reactor Institute, Delft, in 1981, where he worked in the field of neutron diffraction and instrumental neutron activation analysis. In 1986, he joined the Delft Institute of Microelectronics and Submicron Technology (DIMES), Delft. From 1989 to 1998, he was responsible for the mask fabrication and lithography in the DIMES integrated circuit process research sector. During that period he assisted in different research programs. In 1998, he developed a lithography course Applied I-line lithography, two years later the course integrated circuit fabrication technology. These courses, both full week training for engineers from relevant fields in the industry are held for more than 40 times and are also transferred to Tshinghua University, Beijing, China. Besides these educational activities, he cooperated in different research projects related to lithography and microelectromechanical systems (MEMS). He is currently a Senior Researcher with DIMES and involved in different research projects related to MEMS and 3-D integration. He was recently appointed a Lecturer and is author or co-author of more than 60 technical papers.

Angelo Cervone is Assistant Professor in the Aerospace Engineering Faculty at Delft University of Technology. He holds a PhD in aerospace propulsion from University of Pisa (Italy), followed by a 2-years post-doc fellowship at Osaka University (Japan). His main research areas are micro-propulsion, small satellites, systems engineering, cavitation and flow instabilities in liquid rocket engine turbopumps. He is author or co-author of more than 100 papers published in peer-reviewed journals or presented at international conferences. He is member of the Space Propulsion Technical Committee of the International Astronautical Federation, and serves as chairman for several international conferences related to space propulsion.

Eberhard Gill born 1961 in Germany, received a diploma in physics and holds a Ph.D. degree in theoretical astrophysics from the Eberhard-Karls-University, Tuebingen, Germany. He worked as researcher at the German Aerospace Center (DLR) from 1989 to 2006 in the field of precise satellite orbit determination, autonomous navigation, and spacecraft formation flying. He has developed a GPS-based onboard navigation system for the BIRD microsatellite. Dr. Gill has been co-investigator on several international missions, including Mars94-96, Mars-Express, Rosetta, Equator-S, and Champ and acted as principal investigator on the PRISMA formation flying satellite mission. Since 2007, he has held the chair of Space Systems Engineering at the Faculty of Aerospace Engineering of the Delft University of Technology. The research interests of Dr. Gill are spacecraft autonomy, formation flying, Global Navigation Satellite System (GNSS) technology, systems engineering, and combined communication and navigation applications. In 2009, he was been appointed corresponding member of the International Academy of Astronautics (IAA). Professor Gill has also served as the director of the TU Delft Space Institute since its inception in 2015.